

Name of Student: Prajwal Agrawal

PRN: 25070122136

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Subject: Programming with Java

ASSIGNMENT No: 2

TITLE: Constructors in Java

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

```
J ConstructorDemo.java
1  class Student {
2      String name;
3      int rollNo;
4
5      // Default Constructor
6      Student() {
7          name = "Unknown";
8          rollNo = 0;
9      }
10
11     // Parameterized Constructor
12     Student(String name, int rollNo) {
13         this.name = name;
14         this.rollNo = rollNo;
15     }
16
17     void display() {
18         System.out.println("Student Name: " + name);
19         System.out.println("Roll Number : " + rollNo);
20     }
21
22     Run | Debug
23     public static void main(String[] args) {
24         Student s1 = new Student();
25         Student s2 = new Student(name: "Prajwal", rollNo: 101);
26
27         System.out.println(x: "Default Constructor:");
28         s1.display();
29
30         System.out.println(x: "\nParameterized Constructor:");
31         s2.display();
32     }
}
```

```

J ConstructorDemo2.java > ...
1  class MobilePhone {
2      String brand;
3      String model;
4      double price;
5
6      // Default Constructor
7      MobilePhone() {
8          brand = "Samsung";
9          model = "Galaxy A15";
10         price = 20000;
11     }
12
13     // Parameterized Constructor
14     MobilePhone(String brand, String model, double price) {
15         this.brand = brand;
16         this.model = model;
17         this.price = price;
18     }
19
20     // Copy Constructor
21     MobilePhone(MobilePhone m) {
22         brand = m.brand;
23         model = m.model;
24         price = m.price;
25     }
26
27     void display() {
28         System.out.println("Brand : " + brand);
29         System.out.println("Model : " + model);
30         System.out.println("Price : ₹" + price);
31     }
32
33     Run | Debug
34     public static void main(String[] args) {
35
36         MobilePhone m1 = new MobilePhone();
37
38         MobilePhone m2 = new MobilePhone(brand: "Apple", model: "iPhone 16", price: 89999);
39
40         MobilePhone m3 = new MobilePhone(m2);
41
42         System.out.println(x: "Default Constructor:");
43         m1.display();
44
45         System.out.println(x: "\nParameterized Constructor:");
46         m2.display();
47
48         System.out.println(x: "\nCopy Constructor:");
49         m3.display();
50     }

```

Output

```

Default Constructor:
Brand : Samsung
Model : Galaxy A15
Price : ₹20000.0

Parameterized Constructor:
Brand : Apple
Model : iPhone 16
Price : ₹89999.0

Copy Constructor:
Brand : Apple
Model : iPhone 16
Price : ₹89999.0

```

```
Default Constructor:  
Student Name: Unknown  
Roll Number : 0
```

```
Parameterized Constructor:  
Student Name: Prajwal  
Roll Number : 101
```

Conclusion